

Fault Code: 234

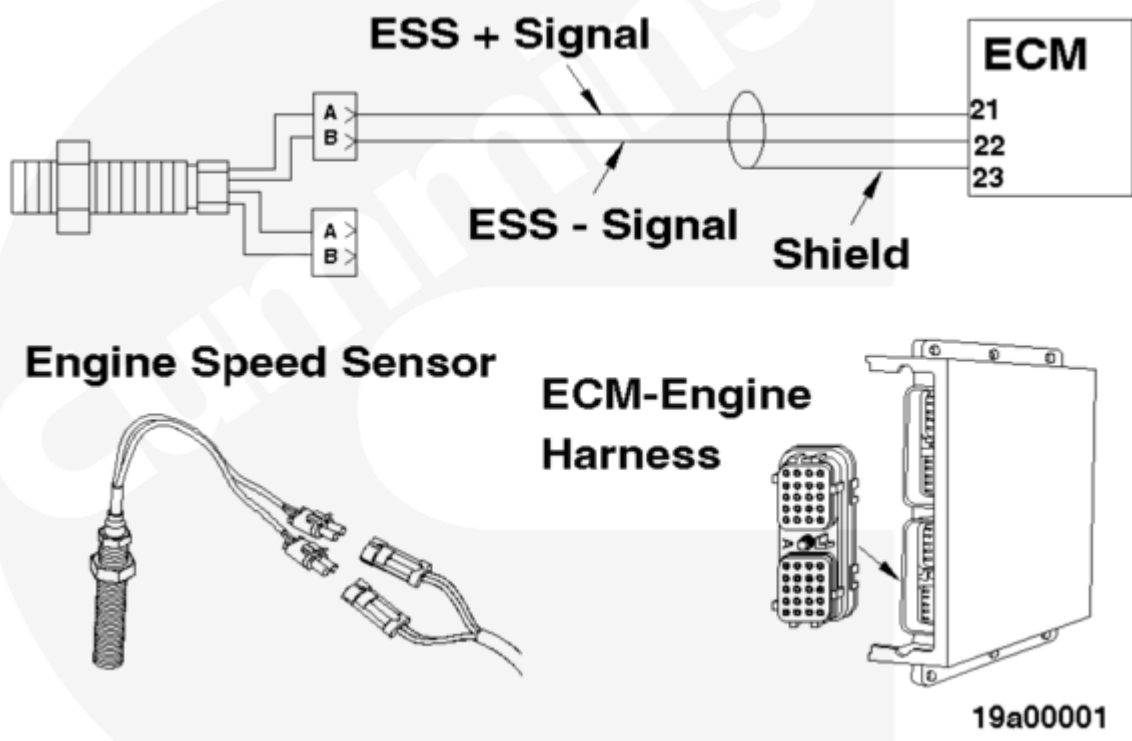
Engine Overspeed - Engine Protection

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Overview

Codes	Reason	Effect
Fault Code: 234 PID(P): SPN: FMI: Lamp: SRT:	Engine Speed Sensor (ESS) signal on pins 21 and 22 of the engine harness electronic control module (ECM) connector indicates engine speed greater than alarm (shutdown) threshold.	Fuel shutoff valves are de-energized (valves closed). Common Alarm output is energized. Overspeed relay driver is energized.

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Circuit Description

The ESS circuit provides the engine speed signal to the ECM through the engine harness.

Component Location

The ESS is located in the flywheel housing.

Shoptalk

This fault code indicates that the engine speed was above the maximum allowable engine speed. An engine overspeed can be caused by either a fuel system problem or the engine being driven or reverse powered past its maximum allowable speed.

The threshold for the engine overspeed shutdown is adjustable with INSITE™, Part No. 3825145. Ensure the threshold is set to the appropriate value.

Troubleshooting Summary

STEPS		SPECIFICATIONS
STEP 1.	Identify the reason for the overspeed.	
	STEP 1A. Check for Fault Code 171.	Fault Code 171 not present
	STEP 1B. Check for motoring of engine (reverse power).	Engine not reverse powered
	STEP 1C. Check for alternate fuel source.	No alternate fuel source
	STEP 1D. Check engine rpm with service tool monitor.	Correct rpm reading
STEP 2.	Clear fault codes.	
	STEP 2A. Disable the fault code.	Fault Code 234 inactive
	STEP 2B. Clear the inactive fault codes.	All faults cleared

STEP 1. Identify the reason for the overspeed.

STEP 1A. Check for Fault Code 171.

Conditions :

- Stop/Run switch in the "STOP" position.
- Controller in the diagnostic mode.

Action	Specification/Repair	Next Step
Read the fault codes using INSITE™, Part No. 3825145.	Fault Code 171 not present	1B
		Go to Fault Code 171

STEP 1B. Check for motoring of engine (reverse power).

Conditions :		
Action	Specification/Repair	Next Step
Check snapshot data for indications of reverse power.	Engine not reverse powered	1C
	Check engine for damage caused by overspeed condition.	2A

STEP 1C. Check for alternate fuel source.

Conditions : - Stop/Run switch in the “STOP” position. - Controller not in the diagnostic mode.		
Action	Specification/Repair	Next Step
	No alternate fuel source	1D
	Locate the alternate fuel source. Locate and remove any alternate fuel source.	2A

STEP 1D. Check engine rpm with service tool monitor.

Conditions : Stop/Run switch in the “RUN” position.		
Action	Specification/Repair	Next Step
Monitor the engine rpm using INSITE™, Part No. 3825145.	Correct rpm reading	2A
	Inspect engine speed sensor Refer to Procedure 019-042.	2A

STEP 2. Clear fault codes.

STEP 2A. Disable the fault code.

Conditions :

- Connect all components.

Action	Specification/Repair	Next Step
• Connect all components. • Start the engine and idle for one minute. • Verify Fault Code 234 is inactive.	Fault Code 234 inactive.	2B
	Return to troubleshooting steps or contact your local Cummins Authorized Repair Location if all steps have been completed and checked again.	1A

STEP 2B. Clear inactive fault codes.

Conditions :

- Connect all components.

Action	Specification/Repair	Next Step
• Erase the inactive fault codes using INSITE™, Part No. 3825145. NOTE: The datalink connector is located on the right bank of the flywheel housing.	All faults cleared	Repair complete
	Troubleshoot any remaining active fault Faults.	Appropriate troubleshooting chart